

AutoDock Vina: A Practical Tutorial

Preparing inputs, running docking, and checking that the answer is real

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Abstract

A working guide to molecular docking with AutoDock Vina: what the program actually computes, how to turn a PDB and a SMILES string into valid inputs, how to read the output, and — the part most tutorials skip — how to verify that your setup produces correct poses rather than merely running without errors. Code examples use the `caddack.docking` module, but every step maps directly onto plain Vina. The pitfalls in Section 9 are ones measured in practice, with the numbers they cost.

1 What docking is, and what Vina computes

Docking answers a geometric question: *given a protein structure and a small molecule, where and in what orientation does the molecule bind?* A docking program has two parts:

1. a **search** over the ligand's position, orientation and flexible torsions, and
2. a **scoring function** that ranks candidate poses.

Vina's scoring function is empirical and pairwise. For every pair of heavy atoms within a cutoff it sums terms depending only on the surface distance $d_{ij} = r_{ij} - R_i - R_j$:

Term	Role
gauss1, gauss2	steric attraction — rewards shape complementarity
repulsion	penalises overlap ($d_{ij} < 0$)
hydrophobic	rewards apolar carbon contact
hydrogen bonding	rewards donor-acceptor pairs at short range
N_{rot} penalty	entropic cost of freezing rotatable bonds

The reported affinity is in kcal/mol; more negative means predicted stronger binding.

What Vina does *not* use

Two omissions matter for how you prepare inputs.

Partial charges are ignored. The Vina scoring function has no electrostatic term — everything above depends on *atom types* and distances. The charge column in a PDBQT file is read but unused. Charges only matter if you switch to the AutoDock4 scoring function (`sf_name="ad4"`), which does have an electrostatic term.

Ring conformations are never changed. Vina samples translation, rotation and *acyclic* torsions only. Whatever ring pucker your input conformer has is what you get. Section 9 shows what this costs when ignored.

2 The PDBQT format

Vina does not read PDB. It reads **PDBQT**: a PDB with two extra pieces of information — a partial charge (Q) and an AutoDock atom type (T).

```
ATOM 1 N TYR A 20 22.423 16.962 23.645 1.00 0.00 -0.346 N
|<-1-6->|<7-11>| |13-16|17|18-20|21|22|23-26| 31-38 55-60 61-66 71-76 78-79
```

The layout is column-strict — a field off by one character makes Vina reject the file with a parsing error. The two added columns are charge (71–76) and atom type (78–79).

Atom types

Typing is what actually drives the scoring function, so getting it right matters more than the charges do.

Type	Element	Meaning
C	carbon	aliphatic, hydrophobic
A	carbon	aromatic
N	nitrogen	<i>not</i> a hydrogen-bond acceptor
NA	nitrogen	hydrogen-bond <i>acceptor</i>
OA	oxygen	hydrogen-bond acceptor (almost all oxygens)
SA	sulfur	acceptor
HD	hydrogen	polar hydrogen — marks a hydrogen-bond <i>donor</i>
Zn, Mg, ...	metals	coordinating centres

A donor is represented by its attached HD hydrogen, not by the heavy atom. This is why a receptor stripped of hydrogens cannot donate hydrogen bonds at all: if you remove every H and mark every nitrogen NA, you have converted every backbone amide from a donor into an acceptor and inverted much of the site's hydrogen-bonding character.

3 Preparing the ligand

The ligand needs 3D coordinates, hydrogens, correct bond orders, and a torsion tree marking which bonds Vina may rotate.

```
from caddack.docking import ligand_pdbqt_from_smiles

ligand = ligand_pdbqt_from_smiles(
    "CC(=O)Oc1ccccc1C(=O)O", # aspirin
    "ligand.pdbqt",
)
```

Internally this is three steps:

1. **Parse and add hydrogens.** Geometry depends on them, and polar hydrogens become the HD donor markers.

2. **Embed a 3D conformer** (RDKit ETKDG) and relax it with MMFF94, or UFF for atom types MMFF does not parameterise.
3. **Write PDBQT** (Meeko), which assigns Gasteiger charges, decides atom types, and builds the torsion tree.

The torsion tree appears in the file as `ROOT/ENDROOT` around the rigid core with `BRANCH/ENDBRANCH` for each rotatable bond. `TORSDOF` at the end gives the count used in the entropic penalty.

Protonation is a modelling decision

A SMILES string does not fix the protonation state. CC(=O)O is neutral acetic acid; CC(=O)[O-] is acetate. They have different atom types, different hydrogen-bonding behaviour, and dock differently. Decide the state at your target pH before docking — tools such as *Dimorphite-DL* or *Epik* enumerate plausible states.

4 Preparing the receptor

This is the fragile step, because a crystal structure is missing most of what Vina needs: **no hydrogens, no bond orders, no charges, no atom types**.

```
from caddack.docking import receptor_pdbqt_from_pdb

receptor = receptor_pdbqt_from_pdb(
    "protein.pdb", "receptor.pdbqt",
    keep_hetatm=["ZN"], # keep a catalytic zinc; drop everything else
)
```

Cleaning

Remove waters and crystallisation additives (glycerol, sulfate, PEG, DMSO), drop alternate conformations, and keep a single model from an NMR ensemble. Keep cofactors and metals that are part of the binding site — deleting a catalytic zinc removes the strongest anchor in the pocket.

Structural waters that bridge ligand and protein are a genuine judgement call: Vina treats the receptor as rigid and dry, so a retained water is a fixed obstacle, while a deleted one leaves a cavity that may be filled incorrectly.

Typing and charges come from residue templates

Because a PDB gives no bond orders, charges cannot be computed from topology the way they are for a ligand. Instead each residue is matched against a **template library** carrying pre-assigned charges and types (this is what Meeko's `ResidueChemTemplates` provides). Consequences:

- Non-standard residues, covalent modifications and interrupted chains fail to match, and there is no fallback that is remotely as good.
- A *truncated pocket file* is not a valid polymer — template matching needs whole residues in a connected chain. Prepare the complete protein.

Histidine, and why it needs a decision

Histidine has three plausible states at physiological pH, and a crystal structure cannot distinguish them because it has no hydrogens:

Template	Protonated at	Net charge
HIE	N ϵ 2	neutral (common default)
HID	N δ 1	neutral
HIP	both	+1

The choice flips which ring nitrogen is a donor and which is an acceptor, so it directly changes docking. Defaulting every histidine to HIE is a reasonable convention; a histidine coordinating a metal or in a catalytic triad may well need HID or HIP. For a principled assignment use **PDB2PQR with PROPKA** (predicts per-residue pK_a) or **Reduce** (adds hydrogens and optimises the hydrogen-bond network, including Asn/Gln/His flips).

5 Defining the search box

Vina searches only inside an axis-aligned box, given as a centre and a size in ångströms.

```
from caddack.docking import box_from_reference_ligand, box_from_pocket_pdb

box = box_from_reference_ligand("known_ligand.mol2", padding=4.0)
box = box_from_pocket_pdb("pocket.pdb", padding=2.0)
```

- **Known site (re-docking, lead optimisation).** Centre on a co-crystallised ligand and pad by roughly 4 å. Tight boxes search well.
- **Known pocket, no ligand.** Use the pocket residues' bounding box, or a cavity detector such as fpocket.
- **Unknown site (blind docking).** A box covering the whole protein works but searches far more space — raise **exhaustiveness** substantially and expect worse ranking.

The box must comfortably contain the ligand in any orientation: a rule of thumb is the ligand's longest dimension plus about 8 å. Vina warns if the box is very large, because search difficulty grows with volume.

6 Running the docking

```
from caddack.docking import dock_pdbqt, dock_smiles

poses = dock_pdbqt(receptor, ligand, box,
                   exhaustiveness=32, n_poses=9, seed=42)

# or end-to-end from a SMILES string
poses = dock_smiles("protein.pdb", "CC(=O)Oc1ccccc1C(=O)O",
                   reference_ligand="known_ligand.mol2")

for p in poses:
    print(p.rank, p.score) # kcal/mol, best first
```

exhaustiveness controls search effort. Vina's default is 8; 32 is meaningfully more reliable for roughly four times the runtime, and blind docking needs more still. It is the first parameter to raise when results look unstable.

Determinism. Vina is stochastic. A fixed `seed` makes single-threaded runs reproducible, but multi-threaded runs may still vary slightly between executions — so treat a single docking run as one sample, not a measurement.

7 Reading the output

Each pose carries a predicted affinity and two RMSD values relative to the best-scoring pose (a spread measure, *not* accuracy against a crystal structure).

Rank	Affinity (kcal/mol)	Interpretation
1	−10.1	best-scoring pose
2	−9.4	plausible alternative
3	−7.2	clearly worse

Two cautions on the number itself:

- **It is not a measured binding constant.** Vina’s scoring function correlates only modestly with experiment — roughly $r \approx 0.6$ on standard benchmarks. Use it to rank poses of *one* ligand confidently, and to rank *different* ligands only cautiously.
- **Scores drift with ligand size.** Larger molecules make more contacts and score better almost mechanically. Ligand-efficiency measures (score per heavy atom) partly correct for this.

A useful diagnostic is the score gap: if ranks 1 and 2 differ by only 0.2 kcal/mol but sit in different orientations, the ranking is not meaningful and both should be considered.

8 Validating your setup: re-docking

Before trusting docking on a new molecule, check that it reproduces answers you already know. **Re-docking** takes a complex whose crystal structure is solved, removes the ligand, docks it back, and measures RMSD to the crystal pose. Below 2 Å is the conventional success criterion.

```
from caddack.docking import pose_rmsd
from rdkit import Chem

reference = Chem.MolFromMol2File("crystal_ligand.mol2")
print(pose_rmsd(poses[0], reference)) # symmetry-aware, in angstroms
```

RMSD must be **symmetry-aware**: a symmetric ring flipped 180° is the same physical pose, and naive atom-order RMSD would score it as a failure. RDKit’s `CalcRMS` matches atoms by substructure and handles this.

Start from the crystal conformer

Re-docking conventionally uses the deposited ligand conformer as the input. This is not circular: Vina randomises position, orientation and acyclic torsions regardless. What it *does* preserve is the ring conformation — and since Vina cannot change ring puckers, re-generating the conformer instead measures conformer generation and docking together, which is a different and harder question.

Reference numbers

Measured on 30 randomly sampled PDBbind refined-set complexes, all of which docked without error:

Protocol	Top pose < 2 Å	Any of 9 < 2 Å	Median RMSD
Re-embedded ligand, <code>exh</code> = 8	43%	70%	2.82 Å
Crystal conformer, <code>exh</code> = 8	53%	83%	1.50 Å
Crystal conformer, <code>exh</code> = 32	60%	90%	1.47 Å

The gap between the last two columns is informative: the search usually *generates* a near-native pose, and the scoring function fails to rank it first about a third of the time. That is the standard motivation for re-scoring a pose set with a better model.

9 Common pitfalls

1. **Re-generating the ligand conformer and expecting Vina to fix the rings.** It cannot. In the table above this alone cost 10 percentage points. The diagnostic signature is a ligand in the *right place* with the wrong shape — one sugar sat 0.67 Å from the correct position yet scored 4.37 Å RMSD; a macrocycle went from 7.67 Å to 0.38 Å once started from the deposited conformer. If your ligand has flexible or macrocyclic rings, dock several ring conformers.
2. **Leaving exhaustiveness at 8 for anything demanding.** Worth about 7 percentage points here for 4× the runtime.
3. **Preparing a receptor with no polar hydrogens.** If every nitrogen is typed NA and there are no HD atoms, the receptor cannot donate a hydrogen bond anywhere. Proper template-based preparation of one test protein gave 554 HD atoms and a 410:9 split of N versus NA, against a naive writer that produced zero donors.
4. **Feeding a truncated pocket file to a template-based preparation.** Interrupted residues fail template matching. Prepare the whole protein, then restrict the *search box* to the pocket.
5. **Deleting metals and cofactors indiscriminately.** A catalytic zinc is often the anchor point of the whole site.
6. **Ignoring protonation.** Both a ligand carboxylate and a receptor histidine change hydrogen-bonding character depending on state, and neither is determined by the input file.
7. **Reading the affinity as a measured ΔG .** It is an empirical score with modest experimental correlation.
8. **Trusting a single run.** Vina is stochastic, and multi-threaded runs are not bitwise reproducible even with a fixed seed. Repeat, or raise `exhaustiveness`, before concluding anything from a small difference.
9. **Judging a docking setup by whether it crashes.** Every protocol in the reference table ran cleanly on all 30 complexes while ranging from 43% to 60% accuracy. Only re-docking distinguishes them.

10 Where to go next

- **Re-scoring.** Because Vina often generates a good pose but ranks it poorly, re-scoring its pose set with a learned model is one of the highest-value additions to a docking pipeline.
- **Flexible side chains.** Vina supports selected flexible receptor residues, at a substantial cost in search difficulty.
- **Better charges.** Only relevant if you move to AutoDock4 scoring or to free-energy methods: the quality ladder runs Gasteiger → MMFF94 → AM1-BCC → RESP.
- **Ensemble docking.** Docking into several receptor conformations partially compensates for Vina's rigid receptor.
- **Virtual screening.** Vina scores are weak at ranking *across* ligands, so screening campaigns normally re-score or re-rank the top slice with a more expensive method.

Vina: Trott & Olson, *J. Comput. Chem.* 31:455 (2010). Vina 1.2: Eberhardt *et al.*, *J. Chem. Inf. Model.* 61:3891 (2021). Meeko and the AutoDock suite: <https://autodock-vina.readthedocs.io>.